



Society of Petroleum Engineers

SPE-226989-MS

Boundary-Integral Type Neural Network (BINN) for Flow Problems in Homogeneous Reservoirs

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/226989-MS](https://doi.org/10.2118/226989-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

This study pioneers the application of Boundary-Integral Neural Networks (BINNs) to subsurface flow simulation, addressing steady-state single-phase flow governed by Laplace-type equations in hydrocarbon reservoirs. BINNs synergistically integrate boundary integral equations (BIEs) with deep learning to overcome limitations of traditional mesh-based methods and Physics-Informed Neural Networks (PINNs). By leveraging Green's functions, BINNs transform partial differential equations into boundary-only formulations, where neural networks exclusively approximate boundary unknowns (pressure/flux), and interior solutions are reconstructed analytically. This approach achieves intrinsic dimensionality reduction, eliminating spatial domain discretization while ensuring mathematical consistency through exact boundary condition enforcement. Numerical validation demonstrates BINNs' computational advantages: In a rectangular reservoir, BINNs achieve sub-0.0002% relative error (9.59×10^{-6} MPa) at interior points. For a complex trapezoidal reservoir with geometric singularities, BINNs attain Boundary Element Method (BEM)-comparable accuracy (max error ~ 0.051 MPa) without specialized singularity treatments. The method's efficiency is evidenced by rapid convergence with minimal boundary sampling and moderate network sizes. BINNs provide a robust, meshless paradigm for reservoir-scale simulation, effectively handling irregular geometries while maintaining high fidelity and scalability, which uncovers its remarkable computational capabilities in the realm of single-phase flow, and offers an inaugural investigation and benchmark for its prospective extensive utilization in numerical reservoir simulations.

Keywords: Deep Learning, Boundary integral equations (BIEs), Physics-informed neural networks (PINNs), Boundary-Integral Neural Networks (BINNs), Meshless Reservoir Simulation, Dimensionality Reduction

Introduction

As the demand for increased accuracy and efficiency in reservoir flow simulation during hydrocarbon exploration and development continues to grow, traditional numerical methods such as the Finite Element Method (FEM) and Finite Volume Method (FVM) face significant challenges in solving the partial differential equations (PDEs) that describe subsurface flow processes. These approaches rely on complex spatial discretization meshes, which pose difficulties when handling intricate geometries or highly nonlinear problems, often resulting in substantial computational costs and mesh generation complexities, while also being prone to numerical dissipation issues (Hughes, 2003; LeVeque, 2002; Rao et al., 2024a).

Recently, deep learning frameworks exemplified by Physics-Informed Neural Networks (PINNs) have emerged, embedding physical laws as residuals within loss functions to enable meshless PDE solutions (Raissi et al., 2019). Leveraging the function approximation capabilities of neural networks, PINNs directly learn the solution mappings from data and governing equations, theoretically circumventing the dependency on spatial discretization meshes characteristic of traditional methods (Liu et al., 2025).

However, PINNs encounter limitations when applied to reservoir-scale flow problems. Their global sampling strategies lead to exponential increases in training costs as the computational domain expands, and they often suffer from reduced accuracy near boundary singularities, such as wellbore regions (Krishnapriyan et al., 2021; Wang et al., 2021). To address these issues, Boundary Integral Neural Networks (BINNs) have been developed (Lin et al., 2023; Sun et al., 2023). BINNs integrate the mathematical core of Boundary Integral Equations (BIE) with deep learning architectures, transforming PDEs into boundary-only integral equations via Green's functions (Brebbia et al., 2012). Within this framework, neural networks are tasked solely with learning boundary unknowns—such as potential functions or their normal derivatives—while the solution within the domain is reconstructed through boundary integrals.

Compared to the global sampling-based solution paradigm of PINNs, BINNs demonstrate distinct advantages in efficiency and robustness: 1) BINNs achieve dimensionality reduction by solving BIEs, requiring only boundary sampling and thus significantly enhancing scalability for large-scale reservoir simulations; 2) they impose boundary conditions analytically, overcoming the approximation errors associated with PINN penalty methods and ensuring mathematical consistency of the solution; 3) the inherent Green's function kernel in boundary integral equations naturally adapts to fundamental solution characteristics, enabling accurate capture of solution behavior in regions with singularities, such as well points or fracture tips, without specialized treatment.

In recent years, substantial progress has been made in the theoretical development and practical applications of BINNs. Lin et al. (2023) introduced the initial BINet framework, combining boundary integral equations with deep learning, utilizing potential theory to convert PDEs into boundary integral forms, and proving network convergence via neural tangent kernel theory. Zhang et al. (2022) pioneered the integration of NURBS boundary parameterization and Telles' singularity treatment strategies into BINNs, validating their effectiveness on Laplace equation benchmarks. Sun et al. (2023) systematically demonstrated the dimensionality reduction benefits of boundary integral-based BINNs in transferring unknowns from the domain to the boundary, simplifying loss function design and improving computational efficiency. Concurrently, operator learning approaches driven by boundary integral equations have advanced, exemplified by Meng et al. (2024), who developed BI-DeepO Net and BI-TDO Net models capable of cross-domain generalization, enabling rapid inference of Laplace and acoustic scattering problems with a single training session. Dai et al. (2024) proposed the BI-Green Net model, which extends the dimensionality reduction concept to Navier-Stokes equations through a boundary integral weight-sharing reconstruction mechanism. The application scope of BINNs has since expanded to elasticity and acoustics. Damiano and van Water school (2025) employed the Physically Informed Boundary Integral Network (PIBI-Net), a variant of BINN, for acoustic field reconstruction, achieving high robustness with boundary-only data. Zhang et al. (2024) first applied BINNs to two-dimensional elasticity and piezoelectric

problems, demonstrating that boundary discretization alone suffices to ensure the automatic satisfaction of governing equations.

Despite these advances, the application of BINNs to subsurface flow modeling remains unexplored. This study aims to investigate the foundational application of BINNs in two-dimensional single-phase reservoir flow problems, constructing a boundary integral-based neural network model for flow simulation, thereby providing a novel methodological approach for numerical modeling in hydrocarbon reservoirs.

Methodology

The governing equation

The seepage governing equations in subsurface porous media primarily consists of the fluid mass conservation equation and Darcy's law, which is derived from physical experiments. Assuming steady flow without source or sink terms, the mass conservation equation for the fluid is given by:

$$\nabla \cdot \mathbf{v} = 0 \quad (1)$$

where $\mathbf{v}$ is the seepage velocity of the fluid, and in two dimensions, $\mathbf{v} = (v_x, v_y)^T$.

Darcy's law is described as:

$$\mathbf{v} = -K \nabla p \quad (2)$$

where $K = \frac{k}{\mu}$ is the hydraulic conductivity (assumed to be a constant scalar, isotropic), p is the fluid pressure.

Substituting Eq. (2) into Eq. (1) yields an equation solely in terms of pressure p :

$$\nabla \cdot (K \nabla p(x)) = 0, \quad x \in \Omega \subset \mathbb{R}^2 \quad (3)$$

In the absence of internal sources or sinks, as shown in Eq. (4), this seepage problem generally has both Dirichlet boundary conditions representing constant pressure boundaries and Neumann boundary conditions representing sealed or constant flow rates (Rao et al., 2022; Rao et al., 2024b).

$$\begin{aligned} \text{Dirichlet boundary condition: } p(x) &= \bar{p}(x), \quad x \in \Gamma_{DB} \\ \text{Neuman boundary condition: } \mathbf{v}(x) &= -K \nabla \bar{p}(x), \quad x \in \Gamma_{NB} \end{aligned} \quad (4)$$

where $\bar{p}(x)$ and $\nabla \bar{p}(x)$ represents the known pressure and pressure gradient vector at point $\mathbf{x}$, and Γ_{DB} is the Dirichlet boundary, Γ_{NB} is the Neumann boundary.

Boundary integral equations (BIEs)

The fundamental solution of the two-dimensional Laplace equation is:

$$G(\mathbf{x}, \mathbf{a}) = \frac{1}{2\pi} \frac{1}{\ln(r)} \quad (5)$$

where $r = \|\mathbf{x} - \mathbf{a}\|$ is the Euclidean distance between the source point $\mathbf{a}$ and the field point $\mathbf{x}$.

After incorporating the fundamental solution, the governing equations can be transformed into boundary integral form based on Green's second equation:

$$\int_{\Omega} (p \nabla^2 G - G \nabla^2 p) d\Omega = \int_{\Gamma} \left(p \frac{\partial G}{\partial n} - G \frac{\partial p}{\partial n} \right) d\Gamma \quad (6)$$

Substituting $\nabla^2 p = 0$ and $\nabla^2 G = -\delta(\mathbf{x} - \mathbf{a})$, and using the screening property of the Dirac δ -function, we obtain the point expression in the domain:

$$\int_{\Gamma} G(\mathbf{x}, \mathbf{a}) \frac{\partial p}{\partial n_x}(\mathbf{x}) d\Gamma(\mathbf{x}) - \int_{\Gamma} p(\mathbf{x}) d\Gamma \frac{\partial G}{\partial n_x}(\mathbf{x}, \mathbf{a}) d\Gamma(\mathbf{x}) \quad (7)$$

When the observation point ξ is located on the boundary Γ , integral singularity needs to be considered: for the $\int_{\Gamma} G(x, a) \frac{\partial p}{\partial n_x}(x) d\Gamma(x)$, G has weak singularity ($\ln(r)$) the integral exists; for the $\int_{\Gamma} p(x) \frac{\partial G}{\partial n_x}(x, a) d\Gamma(x)$, $\frac{\partial G}{\partial n} = \frac{\partial G}{\partial r} \frac{\partial r}{\partial n} \propto \frac{1}{r}$, which has strong singularity, it is necessary to introduce the Cauchy Principal Value (CPV) integral with geometric coefficients $c(a)$, then we obtain (Rao et al., 2018; Rao et al., 2019):

$$c(a)p(a) + \int_{\Gamma}^{CPV} p(x) \frac{\partial G}{\partial n_x}(x, a) d\Gamma(x) = \int_{\Gamma} G(x, a) \frac{\partial p}{\partial n_x}(x) d\Gamma(x) \quad (8)$$

where,

$$c(a) = \begin{cases} 1 & \text{if } a \in \Omega \text{ (interior point)} \\ \frac{1}{2} & \text{if } a \in \Gamma \text{ and } \Gamma \text{ is smooth at } a \\ \frac{\theta}{\pi} & \text{if } a \in \Gamma \text{ is a corner point in 2D} \end{cases} \quad (9)$$

Discretization of BIEs

We discretize the boundary integral equation into a system of linear equations using constant elements. The boundary Γ is divided into n boundary elements Γ_j ($j = 1, 2, \dots, n$) as shown in Fig. 1. Among these, n_1 elements belong to the boundary Γ_{DB} and n_2 boundary elements belong to the boundary Γ_{NB} . For constant elements, points are located at the midpoints of the boundary elements. The function values and normal derivative values on the boundary are assumed constant over each boundary element and equal to the values at the points.

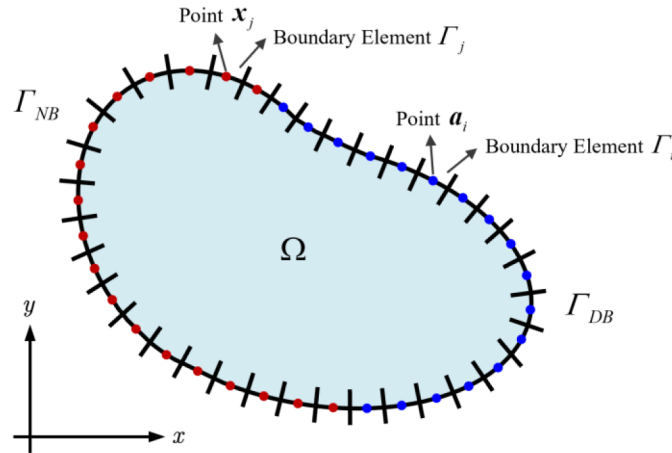


Figure 1—Discretization of computational domain boundary

For point a_i , Eq. (8) can be discretized as:

$$c(a_i)p(a_i) + \sum_{j=1}^n p(x_j) \int_{\Gamma_j} \frac{\partial G}{\partial n_x}(x_j, a_i) d\Gamma(x) = \sum_{j=1}^n \frac{\partial G}{\partial n_x}(x_j) \int_{\Gamma_j} (x_j, a_i) d\Gamma(x) \quad (10)$$

where, denotes the point of boundary elements Γ_i , and x_j represents an arbitrary point within boundary element Γ_j . The integrals over the boundary elements in the equation can be computed a high-order Gaussian quadrature formula.

Define:

$$\begin{aligned}\hat{H}_{ij} &= \int_{\Gamma_j} \frac{\partial G}{\partial n_x}(\mathbf{x}_j, \mathbf{a}_i) d\Gamma(\mathbf{x}) \\ M_{ij} &= \int_{\Gamma_j} G(\mathbf{x}_j, \mathbf{a}_i) d\Gamma(\mathbf{x})\end{aligned}\quad (11)$$

and let $\frac{\partial p}{\partial n_x}(\mathbf{x}_j) = q(\mathbf{x}_j)$, then Eq. (10) can be written as:

$$c(\mathbf{a}_i)p(\mathbf{a}_i) + \sum_{j=1}^n p(\mathbf{x}_j)\hat{H}_{ij} = \sum_{j=1}^n p(\mathbf{x}_j)M_{ij} \quad (12)$$

where,

$$H_{ij} = \hat{H}_{ij} + \frac{1}{2}\delta_{ij} \quad (13)$$

where, δ_{ij} is Kronecker function, that is:

$$\delta_{ij} = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases} \quad (14)$$

So, Eq. (12) can be further simplified to:

$$\sum_{j=1}^n p(\mathbf{x}_j)H_{ij} = \sum_{j=1}^n p(\mathbf{x}_j)M_{ij} \quad (15)$$

Since $\mathbf{n}_1$ function values and $\mathbf{n}_2$ normal derivative values have been prescribed as boundary conditions, solving Eq. (15) therefore yields all unknown quantities on the boundary: specifically, $\mathbf{n}_2$ function values and $\mathbf{n}_1$ normal derivative values.

Neural networks and loss function of the BINNs

BINNs adopt a fully-connected multi-layer perceptron (MLP) architecture, as illustrated in Fig. 2, to approximate the unknown boundary variables in boundary integral equations (BIEs). The network structure consists of an input layer, L hidden layers, and an output layer. Each hidden layer contains $\mathbf{n}_l$ neurons. The input layer takes the midpoint coordinates of boundary elements $\mathbf{x} = (x, y)$, and the output layer generates the pressure p over all boundary elements.

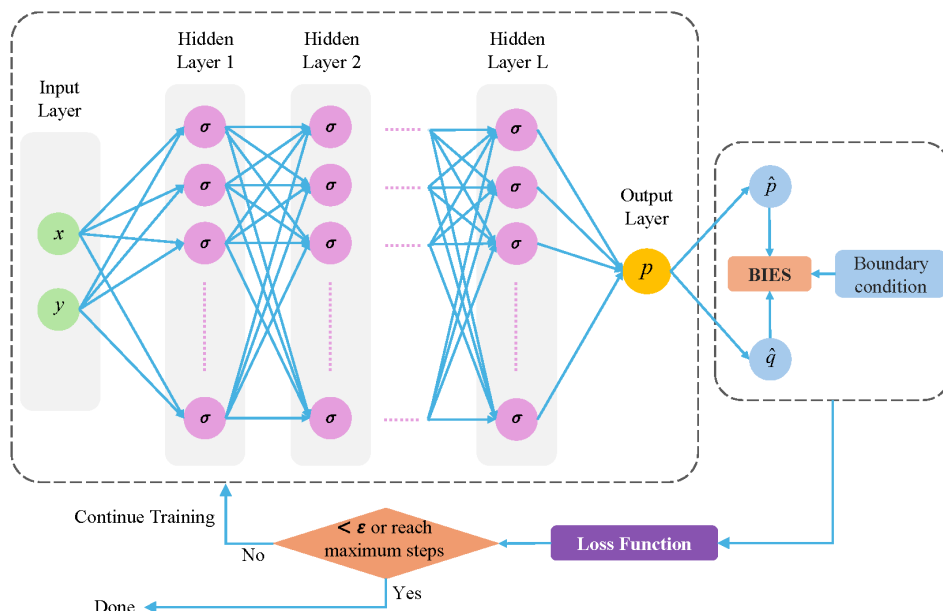


Figure 2—The framework of the boundary integrated neural networks (BINNs).

Leveraging automatic differentiation, the unknown normal derivative $\hat{q}$ on boundary elements is computed from $\mathbf{p}$. Specifically, by applying differentiation operations to $\mathbf{p}$ (with the boundary type determining the differentiation direction, e.g., normal vector orientation), $\hat{q}$ captures the flux-like information consistent with the boundary's geometric and physical constraints. Here, $\mathbf{p}$ inherently includes unknown boundary variables $\hat{p}$, which together with $\hat{q}$ form the key quantities to be optimized.

These unknowns $\hat{p}$ and $\hat{q}$, along with known boundary conditions (e.g., Dirichlet/Neumann constraints), are incorporated into the boundary integral equation (BIE). To train the network, a loss function is constructed by measuring the residual of this BIE. Crucially, $\hat{p}$ and $\hat{q}$ are embedded in the network's computational graph, enabling their gradients (with respect to network weights) to propagate back and optimize $\mathbf{p}$, ensuring the output aligns with both the BIE's mathematical structure and physical boundary constraints.

Mathematically, the network mapping is defined as: for a given input $\mathbf{x}$, the output of the l -th hidden layer is:

$$\mathbf{z}_l = (\mathbf{W}_l \mathbf{z}_{l-1} + \mathbf{b}_l), l = 1, 2, \dots, L \quad (16)$$

where $\mathbf{W}_l$ and $\mathbf{b}_l$ are the weight matrix and bias vector of the l -th layer, respectively; σ_l denotes the activation function (e.g., ReLU, tanh). The output layer maps the hidden layer features to the boundary variables:

$$\begin{cases} p(\mathbf{x}; \Theta) = \mathbf{W}_{\text{out}} \mathbf{z}_L + \mathbf{b}_{\text{out}}, & \text{for Neumann boundary} \\ q(\mathbf{x}; \Theta) = \mathbf{W}_{\text{out}} \mathbf{z}_L + \mathbf{b}_{\text{out}}, & \text{for Dirichlet boundary} \end{cases} \quad (17)$$

where $\Theta = \{\mathbf{W}_l, \mathbf{b}_l\}_{l=1}^L$ represents the network parameters.

For collocation points $\mathbf{x}_n$ on the boundary, the residual term is defined as the mean squared error (MSE) as follows:

$$L_{BIE} = \frac{1}{n} \sum_{j=1}^n [LE(\mathbf{x}_j, p(\mathbf{x}_j; \Theta))]^2 \quad (18)$$

where, $LE(\mathbf{x}_j, p(\mathbf{x}_j; \Theta))$ and $RE(\mathbf{x}_j, q(\mathbf{x}_j; \Theta))$ refers to the left term and right term of Eq. (15).

Once the boundary variables $\hat{p}$ and $\hat{q}$ are obtained via the trained network, the pressure at any interior point is reconstructed using the boundary integral formula:

$$p(\mathbf{x}_{\text{internal}}) = \sum_{j=1}^n (q(\mathbf{x}_j) M_{ij} - p(\mathbf{x}_j) H_{ij}) \quad (19)$$

This formula ensures that the internal solution automatically satisfies the governing PDE, leveraging the integral representation from potential theory.

Results and Discussions

This section validates the computational efficacy of BINNs through two examples of increasing complexity: a rectangular reservoir and a trapezoidal reservoir featuring geometric singularities. The numerical experiments quantitatively assess three critical aspects: (1) accuracy of boundary-to-interior field reconstruction, (2) robustness in handling irregular boundaries, and (3) convergence efficiency.

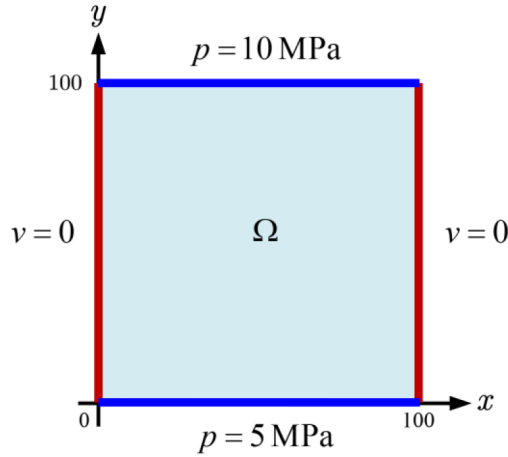
Example 1

Consider a square oil reservoir region $\Omega = [0, 100] \times [0, 100]$ with a side length of 100m. This example investigates the steady-state single-phase seepage flow problem without source or sink terms. The governing equation for this system is the Laplace equation, as follows:

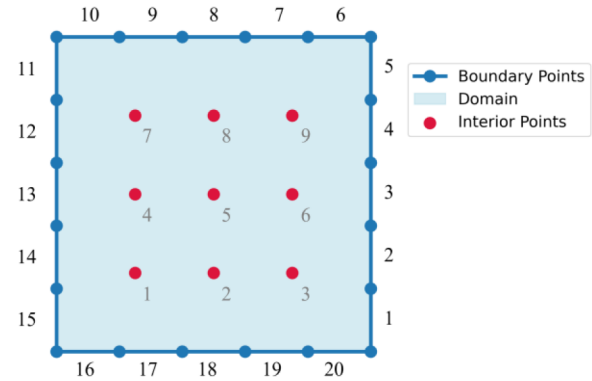
$$\nabla \cdot (\mathbf{K} \nabla p(\mathbf{x})) = 0 \quad (20)$$

where, $K = 10^{-6} \text{m}^2/(\text{MPa} \cdot \text{s})$.

The boundary conditions for this computational domain are defined as follows: the top and bottom boundaries are specified as constant pressure boundaries (Dirichlet conditions), while the left and right boundaries are treated as no-flow boundaries (Neumann conditions), as illustrated in Fig. 3(a).



(a) Computational domain geometry and boundary conditions



(b) Boundary element and interior verification point distribution

Figure 3—The reservoir domain, boundary elements, and Interior points In Example 1.

$$\begin{cases} p(x, 0) = 5 \text{ Mpa}, & 0 \leq x \leq 100 \\ p(x, 100) = 10 \text{ Mpa}, & 0 \leq x \leq 100 \\ v(0, y) = \frac{\partial p}{\partial n}(0, y) = 0, & 0 \leq y \leq 100 \\ v(100, y) = \frac{\partial p}{\partial n}(100, y) = 0, & 0 \leq y \leq 100 \end{cases} \quad (21)$$

As shown in Fig. 3 (b), each of the four edges of the computational domain is discretized into 5 constant boundary elements, resulting in a total of 20 boundary elements. Additionally, 9 interior verification points were selected to validate the accuracy of the numerical solution.

The analytical solution is obtained using the separation of variables method, providing a benchmark for accuracy assessment :

$$p(x, y) = 5 + \frac{y}{20} \text{ MPa}, \quad x, y \in [0, 100] \quad (22)$$

To ensure numerical stability and that the gradients of all parameters during BINN training are of similar magnitude, the coordinates are normalized by scaling them by a factor of 100. This transformation defines new dimensionless coordinates: $x' = x/100$, $y' = y/100$. This coordinate transformation is purely mathematical and does not alter the underlying physical relationships represented by the governing equation. The analytical solution for pressure under the normalized coordinate system (x', y') becomes the scaled version of the original solution:

$$p(x', y') = 5 + 5y' \text{ MPa}, \quad x, y \in [0, 1] \quad (23)$$

For the BINN implementation, the network structure uses a matrix $\mathbf{x}$ composed of the midpoint coordinates of boundary elements as input. The architecture consists of 2 hidden layers, each configured with 10 neurons, employing the hyperbolic tangent (Tanh) function as the activation function. The network training process utilizes the Adam optimizer with a learning rate of 10^{-3} and conducts over 10,000 iterations. Table 1 shows the parameter configuration of Example 1. For pressure calculations at interior points, boundary integral formulas are employed to achieve the complete solution for the entire seepage field.

Table 1—Parameters of the neural networks and optimizers in Example 1

Model	Number of hidden layers	Number of neurons per hidden layer	Total number of parameters	Activation Function	Optimizer (Adams)
BINN	2	10	151	Tanh	10000 iterations, learning rate 10 ⁻³

Fig. 4 demonstrates the convergence behavior of the boundary integral equation (BIE) loss function throughout the training process. The convergence pattern exhibits two distinct phases: during the initial iterations (epochs 0 to 2000), the loss decreases rapidly, indicating that the network parameters quickly adapt to satisfy the boundary integral equation constraints. Subsequently (epochs 2000 to 10000), the loss stabilizes and asymptotically approaches zero, demonstrating that the BINN has successfully fitted both the boundary conditions and governing equations, with the numerical solution accuracy reaching a convergent state.

**Figure 4—Loss function of BIE In Example 1.**

Table 2 presents a comprehensive error comparison between BINN numerical solutions and analytical solutions at the 9 interior verification points. The error distribution across all verification points shows consistent performance, with no significant bias toward any particular region of the domain. The maximum absolute error of 4.32×10^{-3} MPa and the minimum error of 9.59×10^{-6} MPa which represents less than 0.0002% relative error, indicate robust solution quality throughout the computational domain. These results validate the effectiveness of the BINN approach for solving steady-state seepage flow problems in regular geometric domains. The rapid convergence and high accuracy achieved in this example establish a baseline for BINN performance on problems with simple geometries and well-defined boundary conditions. The computational efficiency demonstrated here, combined with the meshless nature of the boundary integral approach, suggests significant potential for extending this methodology to more complex reservoir geometries and boundary condition configurations.

Table 2—Errors of the BINN solution vs. analytical solutions at interior points

Index	Coordinates	BINN solution	Analytical solution	Errors
1	(25, 25)	6.2457	6.25	4.32×10^{-3}
2	(50, 25)	6.2465	6.25	3.51×10^{-3}
3	(75, 25)	6.2460	6.25	3.96×10^{-3}
4	(25, 50)	7.4994	7.50	5.53×10^{-4}

Index	Coordinates	BINN solution	Analytical solution	Errors
5	(50, 50)	7.5000	7.50	9.59×10^{-6}
6	(75, 50)	7.5001	7.50	6.30×10^{-5}
7	(25, 75)	8.7531	8.75	3.10×10^{-3}
8	(50, 75)	8.7543	8.75	4.32×10^{-3}
9	(75, 75)	8.7538	8.75	3.83×10^{-3}

Example 2

This example addresses the oil reservoir seepage flow problem with complex geometric boundaries, investigating the steady -state single-phase seepage characteristics within a trapezoidal oil reservoir region. The computational domain represents a more realistic reservoir geometry with an upper base width of 50m, a lower base width of 100m, and a height of 60m (Fig. 5(a)). The system satisfies the Laplace equation governing conditions without internal source or sink terms.

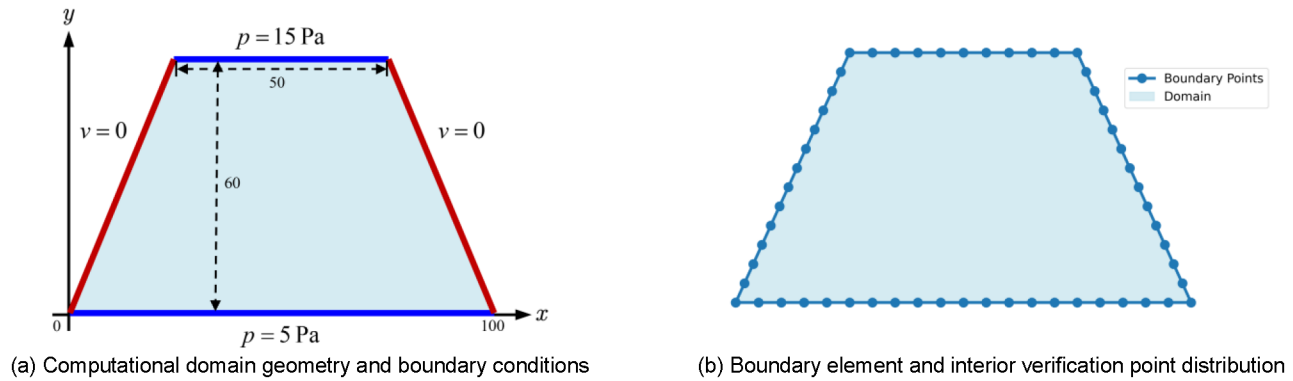


Figure 5—The reservoir domain, boundary elements, and interior points in Example 2

The boundary conditions are configured to simulate realistic reservoir flow patterns: the upper and lower boundaries are specified as constant pressure boundaries (Dirichlet conditions), while the left and right boundaries are treated as no-flow boundaries (Neumann conditions). This configuration creates a pressure-driven flow from the top boundary (15 MPa) to the bottom boundary (5 MPa), with the trapezoidal geometry introducing complex flow patterns due to the converging lateral boundaries (Eq. (24)).

$$\begin{cases}
 p(x, 0) = 5 \text{ MPa}, & 0 \leq x \leq 100 \\
 p(x, 60) = 15 \text{ MPa}, & 25 \leq x \leq 75 \\
 \frac{\partial p}{\partial n}(x, y) = 0, & 0 \leq x \leq 25, 0 \leq y \leq 60 \\
 \frac{\partial p}{\partial n}(x, y) = 0, & 75 \leq x \leq 100, 0 \leq y \leq 60
 \end{cases} \quad (24)$$

The computational domain boundary discretization employs 56 constant boundary elements (each of 5 m length) distributed across all four edges of the trapezoid (Fig. 5(b)).

To address the solution challenges introduced by the complex trapezoidal geometry, the BINN model adopts a significantly enhanced architecture compared to Example 1. The network utilizes a 3-layer hidden layer structure, with each layer configured with 20 neurons, maintaining the hyperbolic tangent (Tanh) function as the activation function. This architectural enhancement, increasing the total parameter count to 921, is necessary to capture the more complex flow patterns and pressure gradients introduced by the irregular boundary geometry. The increased network complexity reflects the fundamental challenge of representing non-uniform flow fields where the pressure gradients vary significantly across the domain due to geometric effects. The additional layers and neurons provide the network with sufficient representational

capacity to approximate the intricate boundary integral relationships inherent in problems with irregular geometries. Table 3 comprehensively summarizes the model parameters.

Table 3—Parameters of the neural networks and optimizers in Example 2

Model	Number of hidden layers	Number of neurons per hidden layer	Total number of parameters	Activation function	Optimizer (Adams)
BINN	3	20	921	Tanh	10000 iterations, learning rate 10^{-3}

The iterative convergence process of the BIE loss function, shown in Fig. 6, demonstrates robust training characteristics despite the increased complexity of the problem. During the initial training phase, the loss value rapidly decreases from approximately 12, and by 900 iterations, the loss approaches zero and stabilizes. This convergence pattern indicates that the enhanced network architecture successfully accommodates the additional complexity while maintaining training stability and achieving satisfactory equation fitting. The convergence rate, while slightly slower than Example 1 due to the increased problem complexity, still demonstrates efficient training characteristics. The stabilization of the loss function near zero confirms that the BINN model has effectively learned to satisfy the boundary integral equation constraints throughout the irregular domain.



Figure 6—Loss function of BIE in Example 2.

Fig. 7 presents a comprehensive comparison of pressure field distributions between the BINN solution and the Boundary Element Method (BEM) reference solution. Both computational approaches demonstrate remarkable consistency in capturing global pressure distribution patterns and local gradient variations. The pressure field characteristics presented by the BEM solution (Fig. 7(a)) align closely with those predicted by the BINN solution (Fig. 7(b)), demonstrating that BINN can effectively handle the mathematical complexities introduced by non-rectangular domain geometries.

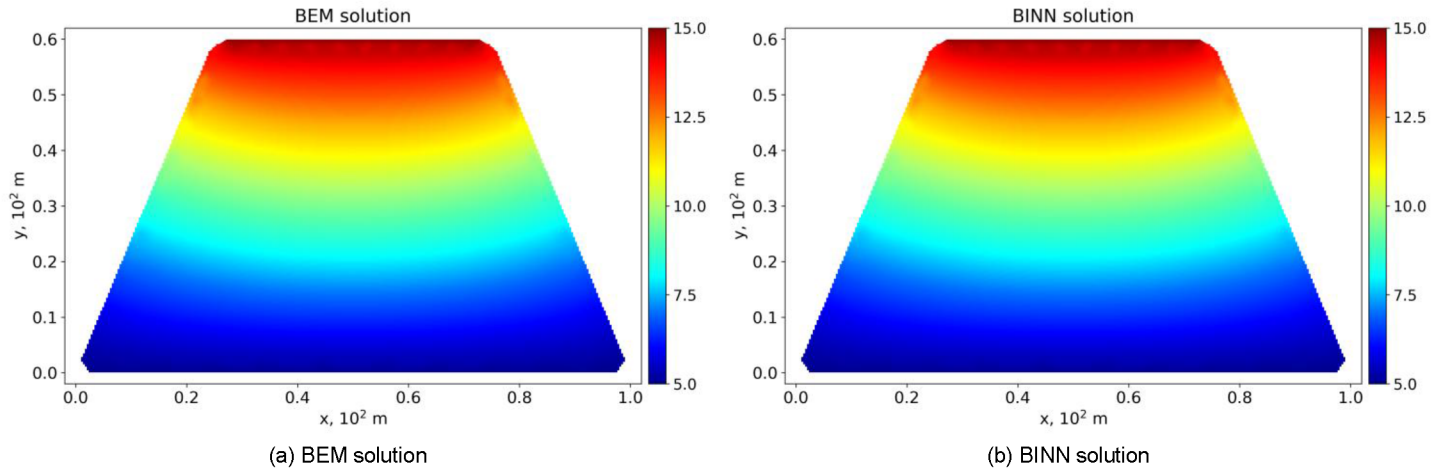


Figure 7—Comparison of pressure field distributions between BINN solution and BEM solution.

The absolute error distribution analysis, presented in Fig. 8, provides detailed insight into the spatial accuracy characteristics of the BINN solution. The maximum error in the entire computational domain is $\sim 0.051 \text{ MPa}$, with an average error of $\sim 0.006 \text{ MPa}$ and a root mean square error of $\sim 0.008 \text{ MPa}$. These error levels represent excellent accuracy for practical engineering applications, particularly considering the geometric complexity of the problem. A notable observation is the localized error concentration phenomenon at both ends of the trapezoidal upper boundary, corresponding to the geometric discontinuity points where the inclined side walls meet the horizontal upper boundary. This error concentration is physically reasonable and mathematically expected, as these locations represent singular points where boundary conditions transition abruptly, creating challenges for any numerical method. Importantly, the errors in the smooth interior regions of the domain approach zero, demonstrating that BINN maintains high solution accuracy away from geometric singularities. The spatial error distribution shows that 99% of the domain exhibits errors below 0.05 MPa , with significant errors confined to less than 5% of the domain area near the geometric discontinuities. The error statistics indicate that BINN achieves numerical accuracy comparable to the well-established BEM under equivalent boundary conditions.

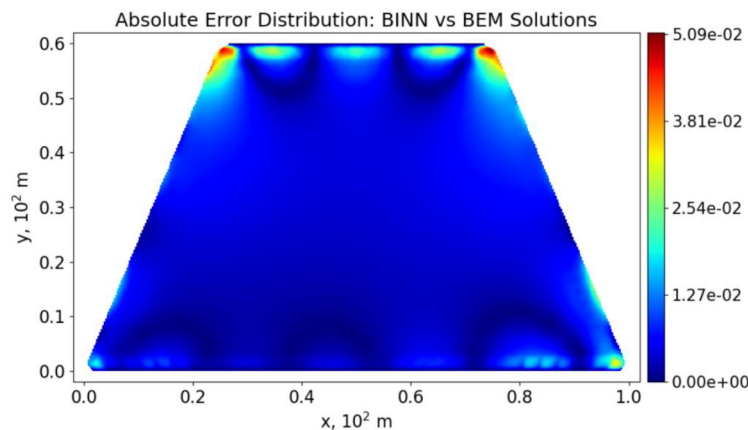


Figure 8—Absolute error distribution of BINN solution and BEM solution (unit: MPa)

The comprehensive comparison demonstrates that BINN achieves numerical accuracy comparable to the BEM through neural network-based global continuous mapping of boundary integral kernels. The method successfully handles irregular geometries while maintaining computational efficiency and solution accuracy. The ability to represent complex flow patterns without traditional mesh generation represents

a significant advantage for reservoir engineering applications where complex geological boundaries are common.

Conclusion

This work establishes Boundary-Integral Neural Networks (BINNs) as a transformative methodology for reservoir flow simulation, demonstrating their efficacy in solving two-dimensional single-phase seepage problems. BINNs fundamentally address key challenges of conventional approaches by unifying BIEs with deep learning: they eliminate spatial discretization meshes required by FEM/FVM and overcome the computational inefficiency of PINNs in large domains through dimensionality reduction. Neural networks learn only boundary variables, while interior solutions are analytically reconstructed via Green's functions, ensuring exact satisfaction of boundary conditions and inherent handling of geometric singularities.

Validation on rectangular and trapezoidal reservoirs confirms BINNs' superior performance. In regular domains, errors as low as $0(10^{-6})$ MPa are achieved with minimal training. For geometrically complex trapezoidal cases with pressure singularities, BINNs match BEM accuracy (~ 0.051 MPa max error) using moderate-sized networks, demonstrating robustness to irregular boundaries. The framework's rapid convergence and boundary-only sampling highlight its computational efficiency and scalability.

BINNs thus offer a paradigm shift toward meshless, boundary-focused reservoir simulation. Future extensions to multiphase flow and fractured reservoirs are warranted, leveraging BINNs' inherent capacity for discontinuous physics, which holds significant potential to advance high-fidelity, computationally efficient forecasting in subsurface engineering.

Acknowledgement

Prof. Dr. Rao acknowledges the support from King Abdullah University of Science and Technology and Yangtze University.

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